

SOCAL WILDFIRES GRID CO-SIMULATION

A validated extreme-event testbed for infrastructure resilience

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Onboarding for new project members

MOTIVATION

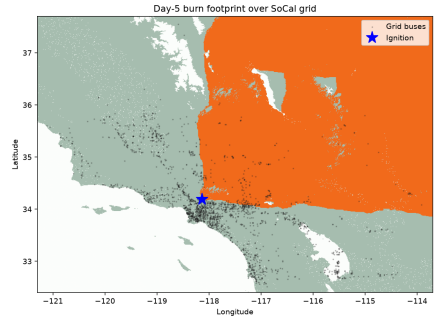
Why This Project Exists

January 2025, Los Angeles basin

- The **Eaton** and **Palisades** fires burned through the LA basin under Santa-Ana forcing.
- Wind-aligned fronts advanced at **60–80 m/min**, with spotting beyond 1 km.
- Distribution circuits, communication links and control systems were overwhelmed *simultaneously*.
- Consequence was not just burned area — it was **loss of electric service** to hundreds of thousands of customers.

The chain this project models

fire spread → asset damage → grid state → resilience metrics



Simulated fire perimeter, day 5
(analysis/fire_perimeter_day5.png)

The Research Gap

Fire models and grid models rarely talk to each other

Fire-centric	Grid-centric	This testbed
Detailed spread physics (Rothermel, Cell2Fire, FARSITE). Infrastructure is <i>absent</i> or a static value layer.	Detailed power-flow and contingency analysis. Fire enters as an <i>exogenous</i> outage list or PSPS scenario.	Validated CA spread <i>and</i> full power-flow re-solve, every step, on one shared geospatial frame . A burned cell and its electrical asset are always co-registered.

- Mao et al. (2026) explicitly call for *feedback-coupled fire/grid simulations*.
- Without coupling you cannot ask: *did that suppression tactic protect the right cells?*

Source: Veith & Sultan, "When Wildfire Breaks the Grid", Sec. 1–2.

GUARDIAN and the Curse of Rarity

The wider research programme

The problem

- DRL controllers are trained on data they have seen.
- Catastrophic events are, by definition, **rare** — they sit outside the training distribution.
- So agents fail *exactly* when resilience matters most. This is the **Curse of Rarity**.

The GUARDIAN answer

- Synthesise physically plausible worst-case worlds from curated GIS data.
- Train foundation policies against them via **Adversarial Resilience Learning (ARL)**.

Why a *synthesised* world?

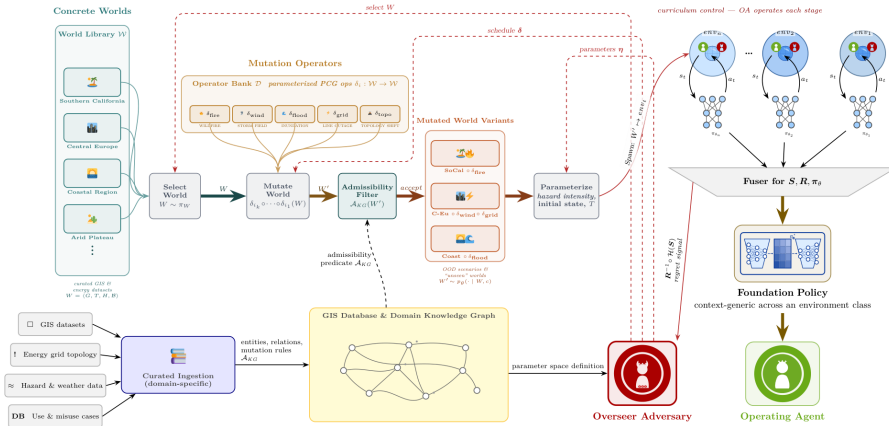
You cannot collect more Eaton Fires. The only way to enlarge the tail of the distribution is to **generate** it — and the only way that generation is defensible is if every mutation stays physically admissible.

Hence the two halves of GUARDIAN: a **mutation operator** that proposes extreme worlds, and an **admissibility filter** that throws away the ones physics forbids.

Motivating precedent: the Feb. 2022 KA-SAT cyberattack disabled 5,800 wind turbines (≈ 11 GW) across Central Europe.

GUARDIAN System Architecture

Where this repository sits in the bigger picture



This repository implements the $SoCal \circ \delta_{fire}$ branch: one concrete world, one mutation operator, one operating agent.

Figure 2 from Sultan, Logemann & Veith, "GUARDIAN".

What You Will Take Away

Learning objectives

1. **Purpose** — what question this testbed is built to answer, and what it deliberately does *not* model.
2. **Concepts** — the constrained-mutation formalism $CMA = (S, \tau, D, \Theta)$ and where learning enters.
3. **Architecture** — how the geospatial, fire, grid and agent layers fit together in palaestrAI.
4. **Navigation** — which directory and which module to open for a given question.
5. **Operation** — the exact commands to reproduce every published figure.
6. **Research** — open ARL questions that are available as thesis topics.

Prerequisite

Basic palaestrAI familiarity. Slide 8 is a short refresher; a one-page reminder of experiment runs, environments and muscles.

PURPOSE AND CONCEPTUAL MODEL

What the Repository Delivers

Four deliverables

1 — NOAA-enriched MIDAS scenario

SoCal co-simulation scenario with weather switched from DWD Bremen to **NOAA HRRR** reanalysis, inside the scenario.

2 — palaestrAI environment

SoCal MIDAS wrapped as a first-class `palaestrai.environment.Environment`, plus an experiment run file.

3 — Wildfire agent (GUARDIAN CMA)

Cellular automaton driven by the Overseer-Adversary parameter vector Θ , over the full SoCal GIS footprint; optional PostGIS layer.

4 — Five-day co-simulation

120 h Santa-Ana wildfire/grid run with KPIs, plots and a generated report.

Source: repository `README.md`.

palaestrAI Refresher

The five nouns you need for this deck

Experiment run A declarative YAML document: environments, agents, phases, termination conditions. Fully reproducible — the document *is* the experiment.

Environment Owns state, publishes **sensors**, accepts **actuators**, is stepped by a simulation controller.

Agent A **Brain** (learns, lives in the Learner process) plus one or more **Muscles** (act, live in RolloutWorkers).

Objective Maps sensor readings to scalar reward. Separate from the environment, so several agents can score the same world differently.

Store Every sensor, actuator and reward is written to a database, so analysis scripts read results instead of re-running simulations.

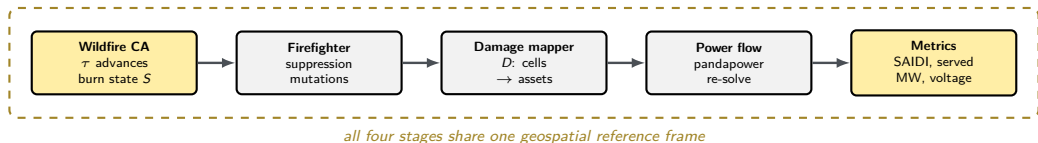
Two conventions used throughout

TakingTurnsSimulationController — agents act in a fixed order, not simultaneously.

DummyBrain — an agent that *acts but never learns*.

End-to-End Simulation Loop

One timestep



- Agents act in **fixed turn order**: wildfire → firefighter → damage_mapper.
- Conflicting cell edits are resolved by *priority*, not by turn order — so the result is deterministic.

Mental Model: $CMA = (S, \tau, D, \Theta)$

The GUARDIAN constrained-mutation automaton

S — state

int8 cell grid, co-registered with the raster.
UNBURNED 0, BURNING 1, BURNED_OUT 2, SUPPRESSED 3,
CONTAINED 5.

τ — transition

The CA sub-step (dt_cma_min, default 5 min). Each burning cell attempts to ignite its 8 Moore neighbours

D — damage map

Fails buses whose cell is burning or burned, and overhead lines inside a radiant-heat clearance buffer.

Θ — the adversary's handle

Ignition point(s), wind vector, dead-fuel moisture, and the global ROS multiplier κ ($1 \leq \kappa \leq 8$).

Θ is the entire attack surface — everything else is physics.

Source: docs/CMA_AGENT.md, wildfire_cma/cma.py.

What Is — and Is Not — Learning

The single most common misreading of this repo

Component	Brain	Learns?
Wildfire (GUARDIAN CMA)	DummyBrain	No — constrained mutation operator driven by Θ
Damage mapper	DummyBrain	No — deterministic consequence model
Firefighter v0.3–v0.5	scripted doctrine	No — expert incident-command heuristics
Firefighter v0.7	FirefighterSacBrain	Yes — SAC with CQL, bootstrapped offline

Why the fire does not learn

This is a deliberate, paper-faithful choice. The published work positions the CA as *the calibrated, validated substrate* on which a learned scenario-generation loop can *later* be built.

Why that matters to you

Turning Θ from a calibrated constant into a learned adversarial policy is the open door into ARL — and the subject of Sec. 6 of this talk.

TECHNICAL ARCHITECTURE

Repository Map

Where to look for what

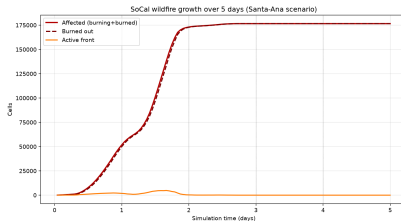
Directory	Contents
wildfire_cma/	The CA itself: <code>cma.py</code> (S, τ, Θ), <code>gis.py</code> , <code>damage.py</code> , <code>wind_field.py</code> , <code>postgis.py</code> . Pure numpy, no <code>palaestraI</code> import.
palaestrai_socal/	Environments (<code>gis_world_env.py</code> , <code>midas_grid_env.py</code>), all agents under <code>agents/</code> , and every <code>experiment*.yaml</code> .
socal_grid/	Geo-referenced pandapower model plus <code>dispatch_and_run.py</code> (the convergence recipe) and <code>build_network.py</code> .
midas_socal/	MIDAS scenario <code>socal_midas.yaml</code> , NOAA weather provider, <code>prepare_midas.py</code> , <code>run_sim.sh</code> .
analysis/	Drivers, report generators, timelapse builders and the committed result figures.
data/	GIS layers, DEM fetcher, CAL FIRE perimeters, offline teacher datasets, CAISO load.
docs/, tests/	Design docs; ≈ 30 test modules (235 passing as of v0.7).

Start reading at `wildfire_cma/cma.py` — it has no framework dependencies.

Geospatial Layer

The shared reference frame

- `SOCAL_BOUNDS = (-121.3, 32.4, -113.7, 37.7)`, EPSG:4326. Row 0 is the *north* edge.
- Three raster sources, in order of fidelity:
 - `from_rasters()` — LANDFIRE FBFM40 fuel + 3DEP elevation (needs `rasterio`).
 - `socal_from_srtm()` — SRTM GL3 (≈ 90 m), 4×3 tile mosaic $\rightarrow 600 \times 760$ grid.
 - `synthetic_socal()` — deterministic CI fallback.
- **Real terrain is the default** — every experiment YAML but the smoke test sets `use_real_dem: true`. The 71 MB mosaic is git-ignored: fetch it once with `fetch_dem_tiles.py`.
- Without it a run *silently* uses synthetic terrain.



Fire growth on the statewide raster
(analysis/fire_growth.png)

Scale trap

One statewide cell ≈ 947 m, so retardant drops are sub-cell — hence fine grids (≈ 50 m) for suppression work.

Wildfire Spread Kernel

Rothermel-derived, wind- and slope-modulated

Rate of spread

$$R(i, j \rightarrow i', j') = \kappa \cdot R^0(i, j) \cdot \varphi_w(u, \theta) \cdot \varphi_s(z, \theta)$$

Ignition probability per sub-step

$$p = 1 - \exp(-R \cdot \Delta t_{\text{CMA}} / \delta)$$

- R^0 — base ROS by fuel class, damped by dead-fuel moisture through the Rothermel coefficient η_m ; zero above FUEL_MX_EXTINCTION.
- $\varphi_w = \exp(c u \cos \alpha)$, $c = 0.25$, tuned so ≈ 20 m/s Santa-Ana wind reproduces the observed **60–80 m/min** fronts.
- $\varphi_s = 1 + 5.275 \tan^2(\text{slope}) \cdot \text{upslope alignment}$.
- δ = cell size in m ($\times \sqrt{2}$ diagonally). A neighbour ignites iff `rng.random()` < p — the only stochastic element, and it is seeded.

#	Fuel class	R^0 m/min
0	non-burnable	0.0
1	grass (GR)	4.0
2	grass-shrub (GS)	2.5
3	shrub / chaparral (SH)	3.2
4	timber-understory (TU)	1.2
5	timber-litter (TL)	0.8
6	slash-blowdown (SB)	1.5

Class 3 dominates SoCal. `gis.fbfm40_to_class()` maps the 40 Scott-Burgan models onto these seven.

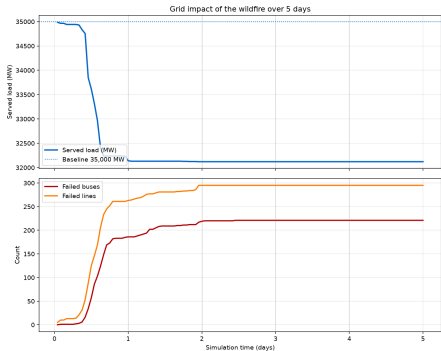
Power Grid Model

The electrical side

- **2,294 buses / 2,595 lines**, geo-referenced pandapower model (socal_grid/socal_grid.json, EPSG:4326, ≈ 5.9 MB).
- Built from open data: OSM substations, transmission lines, power plants, city populations, CAISO load profiles.
- An *illustrative* distribution topology — deliberately **not** a utility feeder model.
- Customers derived via `CUSTOMERS_PER_MW = 200.0`.

Convergence gotcha — read this first

Raw `pp.runpp()` on `socal_grid.json` **diverges**. Always go through `socal_grid/dispatch_and_run.py` — “config D”: strengthen the network, co-locate dispatch, then solve with **Iwamoto Newton-Raphson**.



Grid impact over the five-day run
(analysis/grid_impact.png)

Damage Mapping

From burned cells to de-energised assets

The mechanism

1. Every bus and every line endpoint is projected onto the raster once, at setup.
2. A bus whose cell is BURNING or BURNED_OUT is de-energised.
3. An overhead line is removed if its corridor intersects a burning cell within the *radiant-heat clearance buffer*.
4. Power flow is re-solved; SAIDI and served MW are recorded.

Implementation note

mosaik exposes no `in_service` actuator for buses or lines. So under MIDAS the damage is applied as a **load-shed trip**: write `...load-<bus>-<idx>.p_mw → 0`.

Source: docs/CMA_AGENT.md, docs/AGENTS.md.

Mutation arbitration

Several agents may edit the same cell in one step. A fixed priority makes the outcome independent of turn order:

BURNED_OUT (terminal) > SUPPRESSED > FLOODED
> BURNING > UNBURNED

See `spaces.arbitrate_mutations`. Suppression is not permanent: the environment reverts a cell after `SUPPRESS_PERSIST_STEPS = 12`.

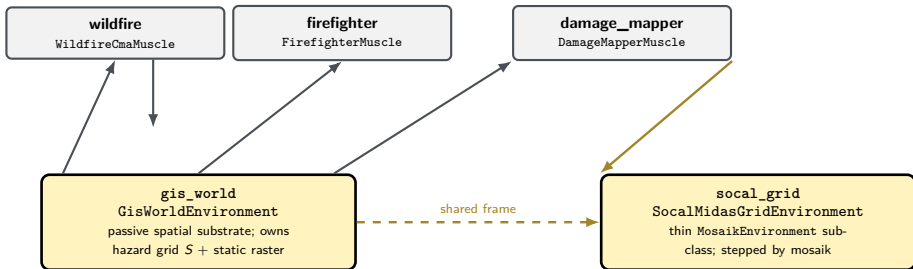
Multi-Agent Architecture

The v0.2 two-environment design

reads `gis.*`
writes `gis.cell_mutations` (BURN-
ING)

writes SUPPRESSED /
LAYER_SUPPRESSION

writes
`...load-<bus>-<idx>.p_mw → 0`



- **gis_world** is **hazard-agnostic**: it steps no dynamics, it only applies `gis.cell_mutations` and re-publishes telemetry. Wildfire today, flood or storm tomorrow.
- Ignition is an *agent parameter* (`ignition_points`, `ignition_step`), **not** an environment actuator.

Firefighting Model Evolution

Five releases, increasing realism



Tunable knobs (v0.4 onward)

`n_planes`, `n_helos`, `n_crews`, `n_dozers`, `n_engines`,
`doctrine` $\in$ {auto, direct, indirect},
`protect_assets`.

Physical constraints are enforced

Fleets degrade above `DEGRADE_WIND_MS` = 13 m/s and are **grounded** at `GROUND_WIND_MS` = 18 m/s — so a 25 m/s high-wind Eaton run is genuinely unchanged by air support.

Source: `CHANGELOG.md`, `agents/firefighter_core.py`.

EXISTING RESULTS

Calibration Against Real Fires

The credibility argument

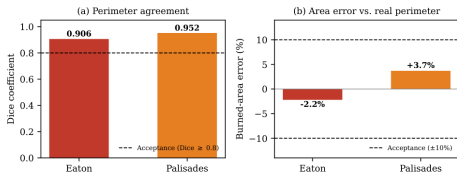


Figure 1 from “When Wildfire Breaks the Grid”.

Fire	Dice	Area err.	Verdict
Eaton	0.906	−2.2 %	PASS
Palisades	0.952	+3.7 %	PASS

- Acceptance bar: $\text{Dice} \geq 0.8$ and $|\text{area \%}| \leq 10$. Both fires clear it.
- Reference: official CAL FIRE perimeters at env step 60. Eaton 13,786 vs 14,102 ac; Palisades 24,677 vs 23,799 ac.

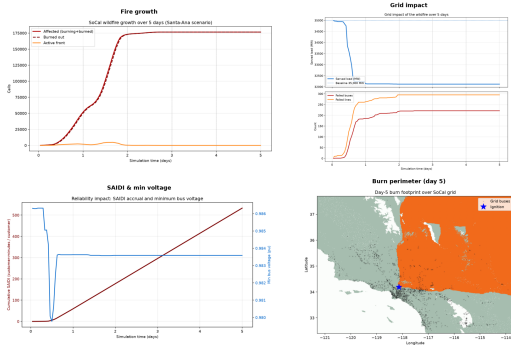
Honest caveat

Without containment the free-running CA *overshoots* by +145 % (Eaton) and +129 % (Palisades). The match relies on `wind_field.py`: perimeter-informed wind, fuel reclassification and `contain_burnable_footprint(..., margin_cells=2)`.

Five-Day Santa-Ana Baseline

The headline scenario

SoCal 5-Day Santa-Ana Wildfire on real SRTM terrain — fire growth, grid impact & reliability



analysis/five_day_combined.png

Setup — 120 hourly steps, seed 47, synthetic 600×760 raster. Strong dry offshore wind on days 1–3, marine-layer recovery on days 4–5.

Outcome

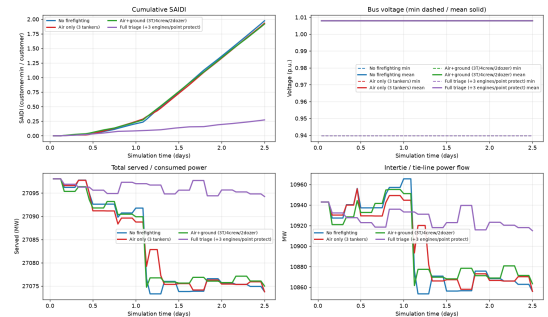
- Served load $\approx 35,000$ MW $\rightarrow$ **31,328 MW**
- ≈ 213 buses and ≈ 327 lines de-energised
- Peak $\approx$ **734,000 customers** disconnected, around day 2.5
- Final burn footprint $\approx 166,000$ cells
- Power flow converged **120/120** steps

Reproduce with `python analysis/run_5day.py --max-steps 120`.

Suppression Counterfactuals

Four phases of incident response

SoCal Eaton Fire — grid impact across firefighter fleets



line | Air only (3 tankers): 18,606 ac (saved 137; -0.05 MW/tanker-hr) | Air+ground (17/14 crew/2 dozer): 18,978 ac (saved 365; 0.01 MW/tanker-hr) | Full triage (1-3 engines/point protect)

analysis/grid_metrics_v04.png

Phase	Acres saved	SAIDI
phase_0_no_ff	—	≈1.97
phase_1_air	137	≈1.95
phase_2_air_ground	365	≈1.93
phase_3_full_triage	654	0.27

The key insight of the whole project

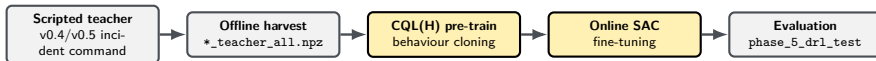
Full triage cuts total burned area by only a few percent — yet it drops SAIDI from ≈1.97 to **0.27** and keeps ≈20 MW more load energised.

A tactic that barely moves total burned area can still be decisive for grid resilience — if it protects the right cells.

On the Eaton fine grid, full triage holds SAIDI 1.61 → 0.24 (≈5,588 ac saved); Palisades ≈1,811 ac saved.

The Learning Firefighter

v0.7 — current frontier



The learning contract

Box(17) observation — burning/burned/suppressed fractions, fire-front geometry, mean slope, wind trigonometry, served MW, SAIDI delta, step progress. *Not* a raster flatten.

Discrete(4) action — NOOP, INDIRECT, DIRECT, TRIAGE. The policy picks a *doctrine*; the same IncidentCommand machinery executes it, gated by resource availability.

Reward — SaidiObjective: $r = -\Delta\text{SAIDI}/\text{scale}$, always ≤ 0 , exactly 0 when load is fully served.

Why gate the actions?

So the learner *cannot invent impossible suppression*. It chooses tactics, not physics.

State this honestly

The v0.7 verification was a **short pipeline smoke run on synthetic-terrain fallback** — not a real-DEM result. Those SAIDI numbers are *not* metric-comparable to production, and two episodes cannot train a policy.

Real-DEM long-runs *are* committed — the two `...drl_long.yml` experiments (episodes: 400, `use_real_dem: true`). They have simply not been run yet. **An open task, not a missing feature.**

Two Bugs Worth Knowing About

Because you will hit their cousins

1 — Ragged sensor arrays

Symptom. `palaestrAI`'s Memory crashed with `ValueError: All arrays must be of the same length`.

Cause. `gis.cell_state` ($\approx 23,660$ elements) was mixed with scalar `*-load-*.p_mw` sensors in one `DataFrame`.

Fix. `agents/_memory_compat.py`, a ragged-safe, idempotent shim for `_MuscleMemory._infos_to_df` — installed in *both* the `RolloutWorker` *and* the `Learner` process.

2 — The silent no-op learner

Symptom. `hARL` logged “could not update”; the firefighter trained for hours *without ever learning*.

Cause. Action-shape mismatch. The muscle emitted `np.array([act_id])` with shape `(1,)`, while the offline loader stored a 0-d scalar. `SACBrain.update()` builds a single `np.array(actions)`, which then went ragged.

Lesson. A DRL agent that silently fails to update looks exactly like a DRL agent that has not converged yet. Assert on your shapes.

Source: `CHANGELOG.md`, v0.7.

RUNNING THE TESTBED

Quick Start

Fifteen minutes to your first figure

1 — Install and verify

```
pip install -r requirements.txt      # or requirements-dev.txt
pytest -m unit                       # < 5 s, numpy only
pytest -m slow                       # loads the 5.9 MB grid, runs power flow
```

2 — Reproduce the headline result (no broker, no database)

```
python analysis/run_5day.py --max-steps 120 --outdir analysis
python analysis/make_timelapse.py    # -> wildfire_timelapse.gif + .mp4
```

3 — Optional: real terrain (free OpenTopography key)

```
export OPENTOPOGRAPHY_API_KEY=...
python data/dem/fetch_dem_tiles.py  # -> socal_srtm_gl3.npz, ~71 MB, git-ignored
```

`run_5day.py` is a standalone driver. It writes `FIVE_DAY_ANALYSIS.md`, four PNGs and `five_day_kpis.csv` — and needs no `palaestraAI` runtime at all.

The Full palaestraI Path

When you need agents, phases and a store

Standard run

```
python midas_socol/prepare_midas.py

palaestrai database-create

# validate the YAML before burning an hour
python -c "from palaestrai.experiment \
import ExperimentRun; \
ExperimentRun.load(
    'palaestrai_socol/experiment.yml')"

palaestrai start palaestrai_socol/experiment.yml
```

MIDAS scenario only

```
midas_socol/run_sim.sh          # full
midas_socol/run_sim.sh --smoke  # CI / quick
```

Multi-phase A/B comparison

```
env PYTHONPATH=$PWD palaestrai \
-c runtime_pg_eaton.conf.yaml start \
palaestrai_socol/experiment_eaton_local_ab.yml
```

--phases grammar

Comma-separated list; each item is phase_uid,
phase_uid:n_planes, or uid:n_planes:Label.
No commas inside a label. Legacy
--phase-a/--phase-b still work.

Pick a runtime*.conf.yaml to match your horizon: runtime_short,
runtime_full16, runtime_full120.

Key Scripts and What They Produce

Your analysis toolbox

Script	Purpose	Output
analysis/run_5day.py	Standalone 120-step baseline run	report + 4 PNGs + KPI CSV
analysis/verify_calibration.py	Assert Dice and area error against CAL FIRE perimeters	pass/fail verdict
analysis/perimeter_validation.py	Compare simulated vs. official perimeter geometry	Dice, area, overlay
analysis/grid_metrics_report.py	SAIDI, voltage, served power across phases	grid_metrics_*.png
analysis/make_timelapse.py	Animate a single run over the Esri satellite basemap	GIF + MP4
analysis/make_comparison_timelapse.py	Side-by-side animation of several phases	GIF + MP4
analysis/make_sim_vs_real.py	Simulated vs. real perimeter figure	comparison PNG
analysis/firefighter_report.py	Scripted-suppression KPI summary	tables + figures
analysis/drl_firefighter_report.py	Learned vs. scripted responder	comparison report

Scripts that read a store take `--store <uri>` and `--phases`; standalone drivers do not. Basemap tiles are cached in `data/basemap_cache/` — no API key needed. Attribution: Esri, Maxar, Earthstar Geographics.

Reproducibility and Provenance

Read before you upgrade anything

Pinned for a reason

palaestrAI constrains `numpy<2` and `pandas==2.1.4`.
Do **not** bump either without re-running the calibration verification — the CA is validated against a specific numerical stack.

Determinism

The only stochastic element is the seeded neighbour-ignition draw. Same seed and same Θ give the same fire, every time.

CI pipeline (`.gitlab-ci.yml`)

lint	flake8, on push
unit	pytest -m unit, on push
system	pytest -m slow, manual
simulate	NOAA smoke + 5-day, manual

simulate publishes its figures as artifacts.

Open data only

USGS 3DEP / SRTM GL3, LANDFIRE FBFM40, OSM, CAISO, NOAA HRRR, CAL FIRE perimeters. GPL-3.0-or-later.

ARL RESEARCH QUESTIONS

From Calibrated Substrate to Adversarial Learning

Where the open work begins

Today

Θ is a **calibrated constant**, set once from the validation values and held fixed. The wildfire agent uses DummyBrain: it acts, it never learns.

The published paper says this explicitly — the testbed is *“the calibrated, validated substrate on which a future learned-scenario-generation loop could be built.”*

The ARL target

Replace the constant with a **learned Overseer-Adversary** that proposes Θ to maximise regret against the operating agent, subject to an *admissibility predicate* $\mathcal{A}_{KG}$ that keeps every proposed world physically plausible.

Both sides then improve together — an autot curriculum in the sense of Baker et al. (2020) and unsupervised environment design.

Why this repository is the right starting point

The hard part — a fire model whose output is *trustworthy* (Dice ≥ 0.9 against two real perimeters) and coupled to a solvable grid — is already done and test-guarded. What is missing is the learning loop around Θ .

Research Questions

Available as thesis topics

RQ1 — Adversarial scenario generation

How can Θ be *learned* rather than calibrated? Which admissibility constraints keep generated worlds physically plausible while still being genuinely hard?

RQ2 — Objective design for resilience

Which signal best represents grid resilience: SAIDI, served MW, fraction of customers connected, or time-to-recovery? How do learned policies differ under each, in the resilience-trapezoid metric space?

RQ3 — Out-of-distribution generalisation

Can a responder trained on Eaton transfer to Palisades, to synthetic fires, or to a different hazard entirely — and how do we measure that honestly?

RQ4 — Reward hacking and simulator artifacts

The suppression action space is deliberately gated. Where else could a policy exploit the model rather than the phenomenon, and how would we detect it?

Research Questions

Continued

RQ5 — The mutation-operator boundary

Where should the line sit between a *constrained mutation operator* (cheap, interpretable, provably admissible) and a *learned adversary* (expressive, but capable of drifting into the implausible)?

RQ6 — Catastrophic forgetting in multi-phase crises

A real crisis is sequential: fire, then outage, then restoration. How do we keep an agent competent at phase 1 while it learns phase 3? (Kirkpatrick et al., 2017.)

RQ7 — Cross-hazard reuse

gis_world is hazard-agnostic by construction. What does it take to add δ_{wind} , δ_{flood} or δ_{grid} from the GUARDIAN operator bank — and do policies trained on fire transfer at all?

A good first paper

Take RQ2. It needs no new physics — only a new Objective subclass and a careful comparison. The store already holds everything you need.

Summary and Your First Contribution

What you now have

- A fire model **validated** against two real 2025 perimeters (Dice 0.906 / 0.952).
- Bidirectional **fire-to-grid coupling** with a full power-flow re-solve each step.
- Four scripted suppression tiers as **counterfactuals**, plus a first learning responder.
- **235 passing tests**, open data only, GPL-3.0-or-later.
- A clear, documented **ARL extension path**.

Pick one, this week

1. **Reproduce.** Run `run_5day.py` and match the published figures.
2. **Read.** Open `wildfire_cma/cma.py` and find equations 6 and 7 in the code.
3. **Perturb.** Change one component of Θ and explain the effect on SAIDI.
4. **Propose.** Write half a page on one of RQ1–RQ7.

Repository

`gitlab.com/ar1-experiments/socal-wildfires`

References and Further Reading

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QUESTIONS?

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